
Shibboleth: Probe-and-Replay Model-Response Watermarking for Diffusion Language Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We propose Shibboleth, a model-response watermarking method for masked dif-
2 fusion language models (DLMs). Unlike an autoregressive sampler, a masked
3 DLM can be re-evaluated on its own output under an arbitrary remasking of the
4 context. In the evaluated DLMs, remasking the context changes the likelihood of
5 an emitted token according to the model weights. These changes define a water-
6 marking channel between the text and the model rather than within token identities.
7 Shibboleth evaluates a keyed family of context probes against each masked block,
8 admits as carriers only the unresolved positions whose block-masked conditional
9 retains mass on both response signs, and resamples at those carriers from the live
10 conditional until the requested sign is committed. Unlike logit-bias methods it
11 leaves the model’s probabilities untouched and conditions only on which model-
12 drawn proposal is accepted; unlike order-steering methods, it does not alter the
13 decoding schedule. Because the admitting gate is invariant to swapping two probes,
14 it stays blind to the sign the key will request, and carriers may therefore be dis-
15 covered adaptively without disturbing the null. Detection re-masks the finished
16 text, rebuilds the identical probes and carriers, and counts keyed sign matches. On
17 text written without the key, the match count is binomial; the resulting p -values
18 are exact at finite sample size, including when carriers are selected adaptively,
19 without a threshold fitted to a calibration corpus. At 1% FPR, Shibboleth reaches
20 0.80 TPR on LLaDA-8B-Instruct and 0.90 on Dream-7B-Instruct. On LLaDA, the
21 token-based detectors reach 0.90–0.93 TPR; Shibboleth’s $0.74\times$ perplexity ratio is
22 the only ratio below its matched unwatermarked baseline in that comparison. On
23 Dream, Shibboleth has the highest measured TPR under settings transferred un-
24 changed from LLaDA. Verification requires 144 masked evaluations per 512-token
25 document. Under random 10% re-denoising, TPR falls from 0.35 to 0.05 at $R = 1$
26 and from 0.88 to 0.50 at $R = 16$; contiguous edits preserve carriers preceding the
27 edit. The method therefore assumes verifier access to the evaluated checkpoint and
28 is not designed for model-free screening.

29 1 Introduction

30 Language-model text can be difficult for readers to distinguish from human writing, creating prove-
31 nance risks in settings that include disinformation, phishing, and academic-integrity review [23, 28].
32 Post-hoc detectors are fitted to a generation process, so their operating characteristics can change
33 when the decoding mechanism changes, including for masked diffusion language models. Water-
34 marking instead embeds a keyed statistical signal during generation and allows a key holder to test a
35 later document [21, 32]. Because a false positive can misattribute human-written text, deployment
36 requires a specified false-positive rate rather than one inferred only from a calibration corpus. Since
37 adversaries also paraphrase output and learn a key’s behaviour from detector queries [19, 23], we
38 study finite-sample control under adaptively selected trials [10, 43].

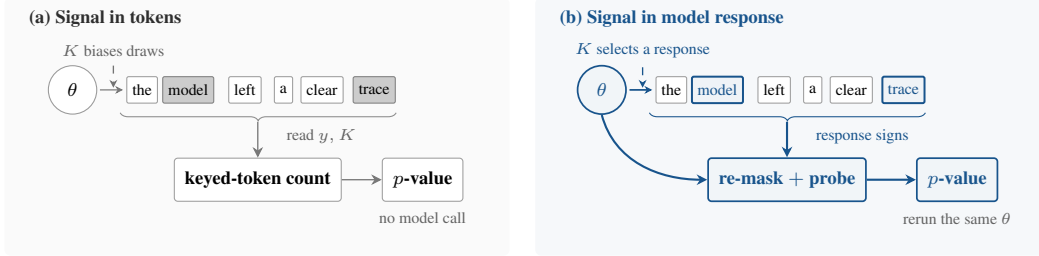


Figure 1: **Token-local and model-response watermarking.** Token-local methods encode the watermark in sampled identities and verify from y, K alone. Shibboleth encodes it in how a fixed checkpoint responds to keyed context probes, so its verifier must rerun that same θ .

Existing watermarks place their secret in the tokens, whether by biasing keyed vocabulary subsets [21], by coupling each draw to pseudorandomness [24], or, in recent diffusion language model (DLM) methods, by exploiting unresolved context and denoising order [17]. Detection then reduces to inspecting the text together with the key, and the generator is never consulted. Masked DLMs, by contrast, permit a different arrangement, since a finished text can be fed back to the model with selected context hidden and the likelihood of an emitted token then moves by an amount the checkpoint alone determines. Hiding a keyed subset of the context is in this sense a *context probe*, the direction in which the likelihood moves is the model’s response to it, and agreement between the keyed sign and that response defines the provenance statistic. We apply a probe where the masked-position conditional assigns nonzero mass to multiple plausible tokens whose responses to two patterns have opposite signs.

Moving the keyed signal from token identities to this relation introduces two requirements absent from token-local detection: the verifier must reproduce the embedder’s probe, and carrier selection must be completed before the accepted answer is known. A gate with access to the rewarded sign would otherwise inflate the false-positive rate above the level the detector reports. Section 3.2 states both as requirements, and existing DLM watermarks avoid them by keeping detection token-local, whether by removing prefix dependence [9], by averaging a red–green bias over unresolved context [14], or by steering which mask is resolved next [17], none of which asks the model anything about the text it produced.

Shibboleth instead meets each condition with one mechanism (Figure 1). It settles replayability by re-masking the current block together with its suffix before scoring the probe family, and the state it measures therefore follows from the final text alone. A *balanced gate* then admits as carriers only the unresolved positions whose block-masked conditional retains mass on both response signs, scoring them in a way that is invariant to swapping the two patterns and therefore blind to the sign the key will ask for. For generation, Shibboleth draws at most R proposals from the unchanged live row at an admitted position and commits the first one carrying that sign, which amounts to rejection sampling and does change the output distribution, although every committed token remains one the model proposed. Verification afterwards re-applies the same probe, reapplies the gate, and returns an exact binomial tail. Unlike a prefix-based watermark, which can require strict left-to-right decoding, and unlike an order-guided one, which changes which mask is resolved next, Shibboleth acts only after the reference schedule has produced a row, and therefore leaves the block partition and the confidence order untouched.

- Across 36 post-hoc substitution settings, the highest-TPR configuration within a $1.35 \times$ perplexity budget reaches 0.10 TPR at 1% FPR; we therefore embed while the conditional is available during generation (Section 3.2).
- We propose Shibboleth, in which block-and-suffix re-masking makes the probe replayable from the finished text and orientation-blind gating keeps the accepted answer independent of how the carriers were discovered, which together keep the detector’s p -values exact without a calibration corpus (Sections 3.3 to 3.5).
- We evaluate Shibboleth on LLaDA-8B-Instruct and Dream-7B-Instruct and report both sides of the trade: $R = 16$, $\kappa = 0.05$ reaches 0.80 and 0.90 TPR at 1% FPR — on LLaDA with the lowest perplexity ratio among the compared rows, and on Dream with the highest TPR under shared

81 settings — whereas verification uses 144 masked evaluations and 10% re-denoising reduces TPR
82 from 0.35 to 0.05 at $R = 1$ and from 0.88 to 0.50 at $R = 16$ (Section 4).

83 2 Related Work

84 **Diffusion language models.** Discrete and masked diffusion objectives carry diffusion [16] over to
85 text [3, 29, 31, 36, 37] and now support instruction-following systems such as LLaDA and Dream
86 [30, 41, 44], with commercial deployments following the same recipe [25]. Because the order in
87 which positions commit affects quality and throughput [2, 6, 20, 38, 39], and because a run can equally
88 be stopped before its step budget is spent [26], accelerators treat the whole schedule as a tunable
89 degree of freedom. Shibboleth, by contrast, treats it as an invariant and draws on masked-token
90 reconstruction only to define a probe it can later repeat.

91 **Watermarking autoregressive text.** Autoregressive watermarks bias keyed token subsets [21] or
92 couple sampling to keyed randomness [1, 13, 24], with later work characterising their reliability
93 and undetectability [10, 22, 43]. Detectors of this family are inexpensive precisely because a key-
94 token hash needs no generator, yet that hash is defined over a resolved prefix and does not transfer
95 unchanged to a context in which most token positions are still masked.

96 **Watermarking diffusion language models.** DLM-specific methods restore a well-defined keyed
97 partition by other means, whether by removing prefix dependence [9], by averaging a red-green
98 bias over unresolved context [14], by keyed sampling [4], or by exploiting the decoding order
99 itself [17, 33, 40]. We compare against dgMARK, an order-steering baseline with a zero-inference
100 detector. Shibboleth differs from all of them in where the watermark signal is stored: dgMARK
101 modifies commit order and reads the outcome from the text, whereas Shibboleth leaves the schedule
102 unchanged and has the verifier rerun the exact checkpoint. Its statistic is therefore bound to model
103 behaviour, and detection requires more than the text and key.

104 **Detector statistics.** Detector designs include alternatives to the asymptotic z -test: exact tails
105 (binomial for green counts, Gamma for Gumbel sums) avoid discrepancies between nominal and
106 empirical FPR caused by Gaussian approximations at small α [12], permutation tests give non-
107 asymptotic p -values for otherwise intractable alignment statistics [24], and windowed maxima must
108 either pay an explicit union-bound correction or be calibrated empirically [22]. When edits leave the
109 watermark signal in only a fraction of the text, sum-based pooling is provably dominated by statistics
110 that concentrate on the surviving fraction [27]. Shibboleth’s binomial tail belongs to the exact-test
111 family, and its block structure admits the same sparse-signal statistics. Under the uniformly random
112 edits in Section 4.2, the pooled statistic has higher measured TPR than the per-block minimum
113 statistic.

114 **Detection that consults the model.** Every DLM watermark above detects from the text and key
115 alone. Other model-dependent or compute-intensive detectors include Kudipudi et al.’s edit-distance
116 detector, which uses quadratic alignment under thousands of permutation resamples [24]; with model
117 access, the Neyman–Pearson score strictly dominates the key-only Gumbel statistic [12], and log-
118 likelihood-ratio detection reads both the watermarked and base conditionals [18]. GaussMark places
119 its key in a Gaussian perturbation of the weights and verifies with one forward-backward pass of the
120 generator [7], and watermarks built to expose model theft assume the verifier can query the suspect
121 system [42]. Unlike GaussMark, which ships a perturbed checkpoint, Shibboleth leaves the released
122 weights as they are and puts the key in the questions asked of them, and one deployed model therefore
123 serves every document key.

124 3 Shibboleth: Watermarking by Model Response

125 3.1 Decoding and Notation

126 Masked diffusion language models differ from autoregressive models in that they predict a missing
127 token from an arbitrary subset of revealed tokens, and the same sequence can consequently be
128 produced under many decoding orders. Let x be a prompt and $y = (y_1, \dots, y_N)$ the continuation to
129 be generated, and for a revealed set $I \subset \{1, \dots, N\}$ with a target position $i \notin I$, the model supplies

130 a predictor $p_\theta(y_i | y_I, x)$ while the remaining positions carry the mask symbol, so that one sequence
131 evaluation returns that distribution at every unresolved position at once.

132 Deployed decoders often restrict that freedom; we study LLaDA’s semi-autoregressive schedule, in
133 which the continuation is partitioned into blocks that finish from left to right while diffusion resolves
134 the masks inside the active block [2, 30], and the reference confidence rule decides which positions
135 commit at each step. Everything below is therefore stated for one block B_b with the prefix F_b that
136 precedes it.

137 3.2 Problem Setup

138 Our goal is a watermark whose detector consults the model rather than the text alone, telling a
139 continuation generated under a secret key from one generated without it at a false-positive rate the
140 verifier states in advance, while preserving the decoding schedule and measuring changes in output
141 quality.

142 **Embedder and verifier.** The embedder receives the model, the prompt, a secret key K , and the
143 target bits, whereas the verifier later receives only the completed sequence, the same model, K , and
144 the public parameters, without logits, random seeds, or commit order. Each document then draws
145 its own key from K and a fresh nonce through a keyed hash, $K_d = \text{HMAC}(K, \text{nonce}_d)$, which to
146 anyone without K behaves as a random function of the nonce [5]. That document key generates both
147 the probe family and the accepted answers, and the false-positive guarantee covers any text produced
148 independently of it.

149 **Threat model.** Both the model and the algorithm may be public, since secrecy rests entirely in
150 K . For every text written independently of the document key, including machine text from another
151 sampler, a detector run at nominal level α fires with probability at most α , and the bound holds at
152 finite sample size and under carriers the method itself selects. We claim nothing in the complementary
153 direction, and an adversary willing to edit the output can therefore suppress detection, as Section 4.2
154 measures. We likewise claim nothing after key disclosure, under the detector-query access that
155 recovers a key’s behaviour [19], or when verification runs a numerically different checkpoint.

156 **Post-hoc substitution.** Post-hoc token substitution avoids intervention during sampling. We mea-
157 sure the number of available replacements under this design. Write τ for the likelihood budget of a
158 substitution, the number of nats by which a replacement may fall below the live argmax. Across 36
159 settings that replace tokens after ordinary generation, the median selected position admits exactly one
160 token at $\tau = 3$, the mean rises from 1.6 at $\tau = 3$ to 5.6 at $\tau = 6$, and the highest TPR inside a $1.35 \times$
161 perplexity budget is 0.10 at 1% FPR. The evaluated method therefore embeds while the conditional
162 is available during generation.

163 We accordingly formalise the problem with three requirements.

- 164 1. **Replayability.** The verifier must be able to put exactly the question the embedder put, from the
165 finished text and the key alone, with no generation trajectory, stored logits, or record of rejected
166 samples.
- 167 2. **Blind selection.** The set of positions that carry a bit may depend arbitrarily on the text and
168 the model, but that choice has to be settled before the sign the key rewards is consulted, since
169 otherwise the reported false-positive rate understates the true one.
- 170 3. **Schedule preservation.** The block partition and the confidence order must survive unchanged,
171 so that a production decoder keeps the latency and quality profile it was tuned for.

172 The first two requirements interact: every scored context must be recoverable from the generated text,
173 while carrier selection must remain independent of the keyed sign. Sections 3.3 to 3.5 specify the
174 corresponding mechanisms.

175 3.3 Probing the Model and Gating the Carriers

176 Unlike dgMARK, which obtains its channel by choosing which mask is resolved next, Shibboleth
177 leaves the decoding schedule as the reference decoder set it. For block B_b , let F_b be the prompt

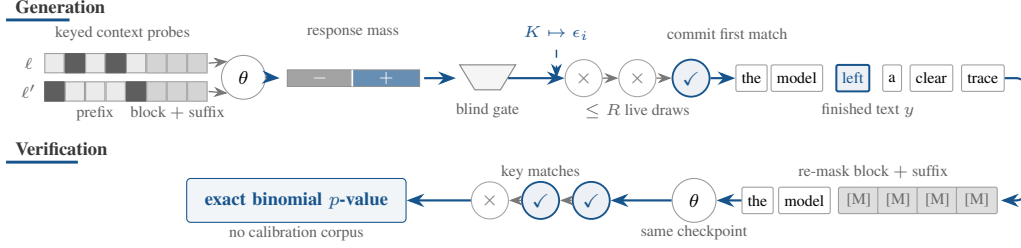


Figure 2: **Shibboleth generation and verification.** Two keyed hidings expose response mass on both signs. A blind gate admits the carrier before the key supplies its orientation, and generation commits the first matching draw. Verification restores the same masked state, replays θ , and returns an exact binomial p -value.

178 together with the completed prefix, and let θ denote the reference checkpoint with p_θ for its predictor.
 179 Shibboleth then constructs L equal-size near, far, and keyed pseudorandom *probe patterns* $D_{b,\ell} \subset F_b$
 180 indexed by $\ell \in \{1, \dots, L\}$, each the set of prefix positions it hides, and it masks $D_{b,\ell}$ together with
 181 the current block and the suffix, and one model evaluation therefore yields, for every token position
 182 $i \in B_b$ and every vocabulary item v , the log-probability

$$\lambda_{i,\ell}(v) = \log p_\theta(v \mid F_b \setminus D_{b,\ell}; B_b \text{ and suffix masked}). \quad (1)$$

183 Probing one block therefore costs $L + 1$ sequence evaluations, one per pattern plus an unprobed
 184 baseline, and the verifier later repeats the same set.

185 Re-masking the block and the suffix removes every token unavailable when the probe was first
 186 applied, which provides replayability: the verifier restores the same state from the final text alone,
 187 and a transient generation-time response becomes a document statistic. Masking only the probed
 188 prefix would not reproduce the embedding state: token positions committed later in the same block
 189 would enter verification-time logits although the future suffix was masked during embedding.

190 For an unordered pair of pattern indices (ℓ, ℓ') and a candidate token z , define the contrast $g_i^{\ell,\ell'}(z) =$
 191 $\lambda_{i,\ell'}(z) - \lambda_{i,\ell}(z)$. Let p_i^{base} denote the block-masked conditional, the model distribution when B_b
 192 and the suffix are masked but no probe pattern is applied. The mass on each sign and the two-sided
 193 score built from them are

$$q_{i,\pm}^{\ell,\ell'} = \sum_z p_i^{\text{base}}(z) \mathbf{1}[\pm g_i^{\ell,\ell'}(z) > 0], \quad S_i^{\ell,\ell'} = 2 \min(q_{i,+}^{\ell,\ell'}, q_{i,-}^{\ell,\ell'}). \quad (2)$$

194 Shibboleth selects, independently at each token position, the pair maximizing $S_i^{\ell,\ell'}$ and stores its
 195 contrast as g_i in the bank $\mathcal{R}_b$, after which the gate admits position i whenever $S_i > s_{\min}$, subject to
 196 an optional per-block cap. Because $S_i^{\ell,\ell'}$ is unchanged when ℓ and ℓ' are swapped, the gate neither
 197 observes nor favours the orientation the key will request, which establishes blind selection, and
 198 both the pair selection and the gating recompute from the block-masked state. Maximizing over
 199 pairs raises the measured mean score from 0.28 for a fixed near/far pair to 0.45, although the score
 200 remains a symmetric proxy computed on the block-masked conditional while generation runs from
 201 an evolving live one.

202 Once the gate has admitted a token position, the key supplies an orientation $\epsilon_i \in \{-1, +1\}$ from
 203 a PRF domain separate from pattern generation, pair selection, carrier assignment, and payload
 204 grouping. A target $w_i \in \{-1, +1\}$ then carries the desired provenance or message bit, so that for a
 205 candidate z the shared embedding-and-verification predicate becomes

$$m_i(z) = \mathbf{1}[w_i \epsilon_i g_i(z) > 0]. \quad (3)$$

206 Should $g_i(z)$ vanish exactly, a second domain-separated fair bit defines $m_i(z)$, and although such
 207 ties cannot be steered, the tie rule keeps the detector's p -values exact.

208 3.4 Embedding the Target Sign

209 Shibboleth inherits the reference confidence order together with the live conditional p_i , and the token
210 positions that the gate rejected are sampled normally. At an admitted carrier define

$$A_i = \left\{ z : w_i \epsilon_i g_i(z) > 0 \text{ and } p_i(z) \geq \kappa \max_v p_i(v) \right\}. \quad (4)$$

211 It draws from p_i up to R times and commits the first member of A_i , and after R misses it commits
212 one fresh unchecked draw instead. Because the row is never recomputed, those retries cost no model
213 evaluation.

214 The resulting sampling distribution can be written explicitly. Let $a_i = p_i(A_i)$ be the acceptance mass
215 and let $q_i = \sum_z p_i(z) m_i(z)$, where $m_i(z)$ applies Eq. (3) and the tie rule to candidate z , so that q_i is
216 the probability that an unconditional draw scores as a detector hit irrespective of the plausibility floor.
217 The committed-token distribution is

$$(1 - (1 - a_i)^R) p_i(\cdot | A_i) + (1 - a_i)^R p_i, \quad (5)$$

218 and its detector-hit probability is

$$\Pr[m_i = 1] = 1 - (1 - a_i)^R (1 - q_i). \quad (6)$$

219 Whenever $\kappa = 0$ and the conditional places no mass on exact ties, $a_i = q_i$ and Eq. (6) collapses to
220 $1 - (1 - q_i)^{R+1}$, so that a_i and q_i diverge only under a floor or a tie rule, where an unchecked fallback
221 may score as a hit without belonging to A_i . The floor meanwhile bounds each guided proposal’s
222 model NLL to within $-\log \kappa$ of the live argmax but constrains neither the fallback nor semantic
223 quality, which is why being model-proposed remains a support statement rather than a distortion-free
224 claim.

225 3.5 Watermark Detection

226 Given a completed text y , the verifier re-masks each block and suffix, rebuilds the probe from the
227 fixed prefix, reapplies the gate, and evaluates $m_i = m_i(y_i)$ at the admitted positions, so that rejected
228 proposals and the generation trajectory are irrelevant. For provenance-only detection, which is all
229 we evaluate here, the admitted carriers form a presence pool P with $w_i = +1$, whereas a payload
230 implementation would keep the group for each payload bit separate, so that a balanced message
231 cannot cancel positive against negative targets. The presence statistic and its one-sided p -value are

$$T = \sum_{i \in P} m_i, \quad p_{\text{det}} = \Pr[\text{Binomial}(|P|, \frac{1}{2}) \geq T]. \quad (7)$$

232 **Proposition 1** (Exact false-positive control). *Assume the PRF domains above behave as independent*
233 *random functions. Fix any text y written without the document key, and condition on the carrier set,*
234 *selected pairs, contrast values, payload roles, and target bits. The orientation bits (and tie bits where*
235 *needed) remain independent fair coins. Consequently the m_i are independent $\text{Bernoulli}(\frac{1}{2})$ variables*
236 *and $T \sim \text{Binomial}(|P|, \frac{1}{2})$ exactly, so that p_{det} in Eq. (7) is an exact p -value and thresholding it at*
237 *α holds the false-positive rate at or below α for every document length and carrier count.*

238 Conditioning fixes the carrier count, selected pair, and contrast magnitude before the independent
239 orientation bit is consulted. Flipping ϵ_i therefore flips m_i whenever $g_i(y_i) \neq 0$, the tie rule supplies
240 a fresh fair bit when it vanishes, and the carriers stay independent of one another throughout. In
241 particular, neither an asymptotic approximation nor a calibration corpus enters anywhere, unlike a
242 z -score whose reference distribution has to be estimated. The guarantee does nevertheless assume
243 text written without the key, together with independence among the several uses of the key, which the
244 domain-separated hash supplies.

245 4 Experiments

246 **Datasets and setup.** We evaluate GSAI-ML/LLaDA-8B-Instruct [30] and Dream-7B-
247 Instruct [41] in fp16 on a single TITAN RTX; Dream emits the prediction for position i at raw-logit
248 index $i-1$, and one shift in the model wrapper restores the shared convention for generation and

Method	Setting	PPL ratio ↓	TPR at FPR ↑		Sequence evaluations ↓	
			1%	0.1%	Generation	Detection
<i>LLaDA-8B-Instruct</i>						
KGW [21]	$\delta = 1$, 512 tok	$1.22\times$	0.90	0.83	512	0
dgMARK [17]	$k = 1$, 512 tok	$1.45\times$	0.83	0.80	512	0
dgMARK [17]	+3-beam, 512 tok	$1.21\times$	0.93	0.93	~2048	0
Shibboleth	$R = 1$, 512 tok	$0.97\times$	0.43	0.20	656	144
Shibboleth	$R = 8, \kappa = 0.1$, 512 tok	$0.87\times$	0.70	0.60	656	144
Shibboleth	$R = 16, \kappa = 0.05$, 512 tok	$0.74\times$	0.80	0.60	656	144
<i>Dream-7B-Instruct</i>						
KGW [21]	$\delta = 1$, 512 tok	$1.84\times$	0.57	0.37	512	0
dgMARK [17]	$k = 1$, 512 tok	$1.75\times$	0.37	0.27	512	0
dgMARK [17]	+3-beam, 512 tok	$0.71\times$	0.30	0.30	~2048	0
Shibboleth	$R = 1$, 512 tok	$1.34\times$	0.40	0.27	656	144
Shibboleth	$R = 8, \kappa = 0.1$, 512 tok	$0.79\times$	0.70	0.67	656	144
Shibboleth	$R = 16, \kappa = 0.05$, 512 tok	$1.09\times$	0.90	0.90	656	144

Table 1: **Watermark detectability.** Detection rate, perplexity against each method’s own matched unwatermarked baseline, and compute in sequence evaluations. All rows are local reproductions on one checkpoint and prompt protocol at 512 tokens; each method keeps its own decoding regime, so PPL ratios are comparable within a method and indicative across methods. A PPL ratio below one does not measure semantic quality. Shibboleth rows use $n = 30$ documents ($n = 20$ for $R = 16$) on both models; Dream PPL ratios rest on the 8–13 pairs per arm that survive the decoded-length rule. Here δ is KGW’s green-list logit bias and k is dgMARK’s candidate count. Every method uses one setting on both models, tuned on neither: the Dream rows carry the LLaDA-chosen hyperparameters, so cross-model rows measure transfer, not each method’s per-model optimum.

replay alike. Detectability is measured on C4 realnewslike [35] prompts truncated to 300 characters as dgMARK’s protocol prescribes. Every row decodes 512 tokens in 32-token blocks at one denoising step per token, each method under its own decoding regime: Shibboleth and dgMARK on the native confidence schedule with full-vocabulary multinomial sampling at temperature 0.8 (dgMARK with top- $k=3$), KGW strictly left-to-right as its green list requires. On top of the reference sampler, Shibboleth adds $L = 8$ probe patterns, threshold $s_{\min} = 0.5$, retry budget R , and an optional floor κ , and we test provenance detection ($w_i = +1$) rather than payload recovery.

Metrics and baselines. Detectability is TPR at analytic FPR thresholds from Eq. (7), and quality is GPT-2-large [34] perplexity against an unwatermarked baseline drawn from the same decoder, reported as $\exp(\text{mean}(\text{NLL}_{\text{wm}} - \text{NLL}_{\text{ctrl}}))$. That ratio measures the distortion each method adds to its own unwatermarked baseline rather than absolute text quality, and comparisons of ratios across methods are therefore indicative rather than controlled. Compute counts sequence evaluations, and each of the $512/32 = 16$ blocks accordingly needs $L + 1 = 9$ of them, hence 144 for verification against $512 + 144 = 656$ for generation. We compare against KGW [21] under left-to-right decoding and dgMARK [17] under confidence decoding.

4.1 Main Results and Analysis

Retry budget and the floor. Detection scales with the number of guided draws as Eq. (6) predicts, since $R = 1$ reaches 0.43 TPR at 1% FPR and raising R from 2 to 4 to 8 on 30 held-out prompts lifts it from 0.73 to 0.83 to 0.87. Retries cost no model evaluation, and the budget is therefore paid in quality instead, with unfloored perplexity ratios climbing from $1.14\times$ at $R = 2$ to $1.40\times$ at $R = 4$ and $1.61\times$ at $R = 8$. The floor converts part of that back, since $(R, \kappa) = (8, 0.1)$ obtains 0.70 TPR at $0.87\times$ and $(16, 0.05)$ obtains 0.80 at $0.74\times$, both below their unwatermarked baselines, although a ratio below one indicates only that GPT-2-large prefers the high-probability tokens the floor admits.

Across models. The cross-model story inverts the LLaDA ranking. Carrying every method’s LLaDA-chosen setting to Dream-7B-Instruct unchanged, both token-local channels weaken sharply: KGW falls to 0.57 TPR at 1% FPR at a $1.84\times$ perplexity ratio, since Dream under forced left-to-right decoding repeats itself heavily (84% duplicate bigrams), and dgMARK falls to 0.37 direct and 0.30 with three-beam lookahead. The model-response channel instead strengthens: the same $(16, 0.05)$ setting reaches 0.90 at both 1% and 0.1% FPR at a sync rate of 0.725 against 0.503 on the control,

Model	Method	Greedy decoding			Multinomial decoding		
		MMLU Acc $\uparrow$	GSM8K Acc $\uparrow$	HumanEval Pass@1 $\uparrow$	MMLU Acc $\uparrow$	GSM8K Acc $\uparrow$	HumanEval Pass@1 $\uparrow$
LLaDA-8B-Instruct	Non-watermarked	0.605	0.740	??	0.595	0.640	??
	KGW [21]	??	??	??	0.535	0.740	??
	dgMARK [17]	??	??	??	0.530	0.660	??
	dgMARK +3-beam	??	??	??	0.530	0.660	??
	Shibboleth	??	??	??	0.575	0.700	??
Dream-7B-Instruct	Non-watermarked	0.740	0.840	??	0.695	0.660	??
	KGW	??	??	??	0.625	0.700	??
	dgMARK	??	??	??	0.570	0.560	??
	dgMARK +3-beam	??	??	??	0.645	0.560	??
	Shibboleth	??	??	??	0.710	0.680	??

Table 2: **Text generation quality.** Accuracy on MMLU [15] and GSM8K [11] and pass@1 on HumanEval [8] under greedy and multinomial decoding, with the unwatermarked run of each model as the reference row of its block. Shibboleth uses $R = 16, \kappa = 0.05$; 50 problems per GSM8K cell, so differences within ± 0.1 are noise. Each watermark decodes under its own regime — KGW left-to-right (its green list requires a resolved prefix; its own control scores 0.640), dgMARK under confidence decoding (its argmax control is the greedy reference cell) — so rows are read against their own controls, not across. Cells marked ?? are pending measured runs.

whose false positives stay at zero, and is on this model the strongest detector in the comparison. A token-local channel depends on how a particular decoder behaves under the constraint the watermark imposes, where a probe the model answers about its own output travels with the model. What does not transfer is the floor’s perplexity dividend, since on Dream (8, 0.1) reads $0.79\times$ but (16, 0.05) reads $1.09\times$ — which (R, κ) is quality-optimal is model-dependent — and the Dream ratios rest on the 8–13 pairs per arm that survive the decoded-length rule, so they carry wide error bars. These are single-setting reproductions, not per-model tuning, for every method alike.

Downstream quality. Perplexity ratios are a proxy, so we additionally measure GSM8K [11] accuracy on 50 test problems at 256 tokens, comparing the watermarked $R = 16, \kappa = 0.05$ sampler against its matched unwatermarked control under the identical decoding schedule (Table 2). The watermarked arm scores 0.700 against the control’s 0.640, and the paired view shows five problems flipping right and two flipping wrong, a tie within noise at this cohort size rather than an improvement; the sub-control perplexity ratio is therefore not purchased with degenerate text. Detection on these short outputs is weak (0.12 at 1% FPR) since 256 tokens admit few carriers, and the control arm sits at its null (0.02 at 1%). The peers behave the same way under their own controls (Table 2): no method moves GSM8K accuracy beyond noise at this cohort size, and short task outputs starve every detector, not only ours — direct dgMARK also reads 0.12 at 1% FPR, its three-beam variant 0.44, and KGW at $\delta = 1$ embeds almost nothing detectable at all (mean $z = +0.46$, zero TPR), since a short, heavily constrained answer leaves its bigram channel few free choices. A watermark’s headline TPR, measured on 512-token continuations, does not transfer to 256-token task outputs for any method here. On Dream the task-quality cost separates the methods more sharply than any perplexity ratio did: KGW leaves accuracy untouched (0.700 against its own control’s 0.700) while barely embedding (0.14 at 1%), whereas dgMARK drops it from 0.840 to 0.560 in both variants — twenty-eight against forty-two problems of fifty, well beyond noise — for 0.18 and 0.66 TPR respectively. Shibboleth holds its Dream control’s accuracy (0.680 against 0.660, nine problems flipping right and eight wrong), so on this model it is simultaneously the strongest 512-token detector in Table 1 and the only watermark here that pays no measurable task accuracy for it. MMLU repeats the pattern at 200 questions per cell: Shibboleth sits within noise of its control on both models (0.575 against 0.595 on LLaDA, 0.710 against 0.695 on Dream), while dgMARK’s direct variant gives up 0.075 on LLaDA and 0.170 on Dream against its own control.

Cost against token-local detectors. Detection trails the token-based detectors at matched length: dgMARK’s three-beam variant reaches 0.93 TPR at both 1% and 0.1% FPR against 0.80 and 0.60 for Shibboleth, its direct variant reaches 0.83 and 0.80, and KGW reaches 0.90 and 0.83. Shibboleth instead carries the only perplexity ratio in the comparison that reads below its own unwatermarked baseline, $0.74\times$ against $1.22\times$ for KGW, $1.45\times$ for direct dgMARK and $1.21\times$ for three-beam dgMARK; every baseline row costs perplexity, whereas the floor discards exactly the low-probability draws a proxy scorer punishes. KGW additionally requires strict left-to-right decoding, under which

Attack	Strength	TPR at FPR $\uparrow$	
		1%	0.1%
<i>Configuration A: $R = 1$, 512 tokens ($n = 20$)</i>			
None	—	0.35	??
Re-denoising (same model, argmax)	10% of positions	0.05	??
Deletion	10% of positions	0.05	??
Insertion	10% of positions	??	??
Substitution (random)	10% of positions	??	??
Copy-paste	25% kept, 1 segment	??	??
Paraphrase (DIPPER) [23]	lex 20, order 0	??	??
<i>Configuration B: $R = 16$, $\kappa = 0.05$, 1024 tokens ($n = 16$)</i>			
None	—	0.88	0.88
Re-denoising (random positions)	5% of positions	0.88	0.75
	10% of positions	0.50	0.38
Re-denoising (contiguous span)	10% of positions	0.88	0.88
	30% of positions	0.88	0.88

Table 3: **Detection after post-editing.** Detection after each attack under the evaluated pooled detector, scored against the same analytic thresholds as Table 1. The signal written through a replayable probe is synchronised to the prefix, so damage propagates forward only: at $R = 1$, a 10% edit invalidates nearly every carrier. At $R = 16$, $\kappa = 0.05$, TPR at 1% FPR is 0.88 after a random 5% edit, 0.50 after a random 10% edit, and 0.88 after a contiguous 30% edit; carriers before the edited span remain unchanged. Small cohorts: one document is 0.05 ($n = 20$) or 0.06 ($n = 16$) of TPR.

315 this checkpoint repeats itself heavily, and its repeated bigram events must be deduplicated: without
316 deduplication its control measured 37% FPR, and even with it two of thirty 512-token null documents
317 cross the 0.1% threshold, a residual calibration failure of exactly the kind an exact conditional null
318 rules out by construction [12]. Shibboleth alone, against all of that, spends 144 masked evaluations
319 to detect at all, although paying for detection is not unprecedented: edit-distance detection runs a
320 quadratic alignment under thousands of permutation resamples [24], and GaussMark verifies with
321 a forward-backward pass of the generator itself [7]. Cohorts of twenty to fifty documents leave a
322 standard error near 0.08 on a TPR of 0.80, and two rows separated by less than about a tenth should
323 accordingly be read as a tie.

324 4.2 Calibration and Robustness

325 **Calibration under the shipped rule.** Proposition 1 is analytical, which is why we also measure
326 the rule as deployed. Across 20 keys and 120 non-watermarked generations apiece on LLaDA, the
327 mean FPR is 0.0121, 0.0037 and 0.0000 at nominal 0.10, 0.05 and 0.01, and the worst individual
328 key gives 0.0583, 0.0250 and 0.0000, and control consequently holds for every key rather than on
329 average alone, which is what binds when one key serves many documents. Every Dream control
330 cohort in this section behaves the same way, with zero false positives at 1% throughout.

331 **Post-editing.** Edit robustness is the channel’s principal failure mode, and it degrades abruptly, since
332 same-model argmax re-denoising of a random 10% of token positions takes TPR at 1% FPR from
333 0.35 to 0.05 on 20 $R = 1$ documents and deleting 10% measures the same. The collapse is one of
334 synchronisation rather than signal strength, because an early edit alters the fixed prefix from which
335 every later probe is rebuilt, and a single change thus invalidates every carrier after it. That cohort
336 uses $R = 1$, and the headline setting at $R = 16$ consequently remains unattacked, which we record
337 as an open gap rather than as evidence of resilience. Redundancy, however, changes the picture:
338 attacking the strongest setting ($R = 16$, $\kappa = 0.05$ at 1024 tokens, $n = 16$) rather than the weakest,
339 re-denoising 5% of positions leaves TPR at 1% FPR unchanged at 0.88 and re-denoising 10% halves
340 it to 0.50 instead of erasing it, with false-positive control holding on the identically attacked controls.
341 We also measured the block-local repair the pooled statistic seems to invite: a per-block exact test
342 combined by Bonferroni over the blocks keeps the document-level guarantee by union bound, yet it
343 never beats pooling here — 0.38 against 0.50 at 1% FPR under the 10% attack, and 0.69 against 0.88
344 clean — because uniformly random edits damage every block, so the sparse-signal regime in which
345 concentration-type tests dominate sum-based ones [27] never arises. Bounding the conditioning
346 window is the corresponding structural repair on the embedding side: with a 128-token window the
347 response bank reads only the prompt and the last 128 generated tokens, so one edit can perturb at most

five blocks’ banks instead of every later one. Measured at $R = 16$, $\kappa = 0.05$ and 512 tokens on shared prompts and seeds ($n = 16$), the window costs nothing clean — 0.69 TPR at 1% FPR in both arms, at a comparable perplexity ratio — but buys nothing under random 10% re-denoising either (0.44 against 0.50, a one-document difference), and for the same reason: the window bounds propagation, while uniformly random edits already damage every window. The localized-edit test then resolves both questions at once, and in the deployed detector’s favour: re-denoising one contiguous span of 10% or even 30% of positions leaves the pooled count exactly at its clean 0.88 at both operating points, because desynchronisation propagates forward only and every carrier before the span survives untouched — what an edit costs is set by where it starts, not by how much it changes. Neither repair is needed in the regimes measured here: Bonferroni-min never overtakes pooling, and the window has nothing left to protect. The one untested regime that could still favour a min-statistic is copy–paste-style sparse survival, where only a fragment of watermarked text remains inside a larger document.

5 Conclusion

Shibboleth computes a provenance statistic from a masked DLM’s responses to keyed probes of its own output rather than from a keyed pattern in token identities. Because carrier selection is completed before the keyed orientation is consulted, adaptive carrier discovery preserves the binomial null. The method also preserves the reference decoding schedule, but verification requires the specified checkpoint; a key–token hash does not have this requirement. Under settings transferred unchanged from LLaDA to Dream, the two token-local baselines have lower detection rates, and dgMARK reduces Dream GSM8K accuracy by 0.28, whereas Shibboleth’s probe is recomputed with the Dream checkpoint. Token-local methods retain lower verification cost and do not require checkpoint-specific replay.

Limitations. That guarantee is nevertheless one-sided, since soundness holds for any text written without the key while nothing is claimed in the other direction, and editing degrades detection: a single same-model denoising pass over a tenth of the positions takes TPR from 0.35 to 0.05 at $R = 1$, and at $R = 16$, $\kappa = 0.05$ it reduces detection from 0.88 to 0.50 at 1% FPR ($n = 16$); contiguous edits leave the measured TPR unchanged even at 30% of positions, since damage propagates forward only. The measured edit results are limited to random and contiguous re-denoising under the reported redundancy settings. Verification requires 144 masked evaluations per document, compared with zero for the token-local detectors in Table 1. The response statistic is bound to one checkpoint as well; different weights or numerical precision fall outside the stated replay assumptions. Finally, the evaluation covers two masked diffusion models, one prompt corpus, cohorts of twenty to fifty documents, perplexity ratios read against each method’s own unwatermarked baseline plus a single 50-problem GSM8K probe, and provenance detection only, with the payload construction of Section 3.5 never measured.

Future work. Content-anchored probes can be evaluated as a method for limiting desynchronisation after insertions or deletions, and copy–paste attacks, where a per-block minimum statistic may differ from pooled counting. Sequential testing can reduce verification work because the binomial tail is monotone in the accumulated match count. Quantised contrasts may reduce sensitivity to checkpoint precision at the cost of carrier capacity. Query-adaptive attacks also require separate evaluation because detector responses reveal key-dependent information. A combined system could use a token-local first stage followed by model-response verification on selected documents.

References

- [1] Scott Aaronson and Hendrik Kirchner. Watermarking GPT outputs, 2023. <https://www.scottaaronson.com/talks/watermark.ppt>.
- [2] Marianne Arriola, Subham Sekhar Sahoo, Aaron Gokaslan, Zhihan Yang, Zhixuan Qi, Jiaqi Han, Justin T. Chiu, and Volodymyr Kuleshov. Block diffusion: Interpolating between autoregressive and diffusion language models. In *ICLR*, 2025.
- [3] Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *NeurIPS*, 2021.
- [4] Arka Bagchi, Akhil Bhimaraju, Moulik Choraria, Daniel Alabi, and Lav R. Varshney. Watermarking discrete diffusion language models. *arXiv preprint arXiv:2511.02083*, 2025.
- [5] Mihir Bellare, Ran Canetti, and Hugo Krawczyk. Keying hash functions for message authentication. In *CRYPTO*, 1996.
- [6] Heli Ben-Hamu, Itai Gat, Daniel Severo, Niklas Nolte, and Brian Karrer. Accelerated sampling from masked diffusion models via entropy bounded unmasking. In *NeurIPS*, 2025.
- [7] Adam Block, Alexander Rakhlin, and Ayush Sekhari. GaussMark: A practical approach for structural watermarking of language models. In *ICML*, 2025.
- [8] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.
- [9] Ruibo Chen, Yihan Wu, Yanshuo Chen, Chenxi Liu, Junfeng Guo, and Heng Huang. A watermark for order-agnostic language models. In *ICLR*, 2025.
- [10] Miranda Christ, Sam Gunn, and Or Zamir. Undetectable watermarks for language models. In *COLT*, 2024.
- [11] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [12] Pierre Fernandez, Antoine Chaffin, Karim Tit, Vivien Chappelier, and Teddy Furon. Three bricks to consolidate watermarks for large language models. In *2023 IEEE International Workshop on Information Forensics and Security (WIFS)*, 2023.
- [13] Jiayi Fu, Xuandong Zhao, Ruihan Yang, Yuansen Zhang, Jiangjie Chen, and Yanghua Xiao. GumbelSoft: Diversified language model watermarking via the GumbelMax-Trick. In *ACL*, 2024.
- [14] Thibaud Gloaguen, Robin Staab, Nikola Jovanović, and Martin Vechev. Watermarking diffusion language models. In *ICLR*, 2026. arXiv:2509.24368.
- [15] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *ICLR*, 2021.
- [16] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020.
- [17] Pyo Min Hong and Albert No. dgMARK: Decoding-Guided Watermarking for Diffusion Language Models. In *ICML*, 2026. arXiv:2601.22985.
- [18] Zhengmian Hu, Lichang Chen, Xidong Wu, Yihan Wu, Hongyang Zhang, and Heng Huang. Unbiased watermark for large language models. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- [19] Nikola Jovanović, Robin Staab, and Martin Vechev. Watermark stealing in large language models. In *ICML*, 2024.

- [20] Jaeyeon Kim, Kulin Shah, Vasilis Kontonis, Sham M. Kakade, and Sitan Chen. Train for the worst, plan for the best: Understanding token ordering in masked diffusions. In *ICML*, 2025.
- [21] John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A watermark for large language models. In *ICML*, 2023.
- [22] John Kirchenbauer, Jonas Geiping, Yuxin Wen, Manli Shu, Khalid Saifullah, Kezhi Kong, Kasun Fernando, Aniruddha Saha, Micah Goldblum, and Tom Goldstein. On the reliability of watermarks for large language models. In *ICLR*, 2024.
- [23] Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John F. Wieting, and Mohit Iyyer. Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense. In *NeurIPS*, 2023.
- [24] Rohith Kuditipudi, John Thickstun, Tatsunori Hashimoto, and Percy Liang. Robust distortion-free watermarks for language models. *TMLR*, 2024.
- [25] Inception Labs, Samar Khanna, Siddhant Kharbanda, Shufan Li, Harshit Varma, Eric Wang, Sarah Birnbaum, Ziyang Luo, Yanis Miraoui, and Akash Palrecha. Mercury: Ultra-fast language models based on diffusion. *arXiv preprint arXiv:2506.17298*, 2025.
- [26] Chia-Ming Lee, Shao-Kai Liu, Ming-Ching Chang, Xin Li, Yu-Lun Liu, and Chih-Chung Hsu. Commit locally, exit globally: Coordinating adaptive sampling and early exit in diffusion language models, 2026. URL <https://arxiv.org/abs/2607.28166>.
- [27] Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, and Weijie J. Su. Robust detection of watermarks for large language models under human edits. *Journal of the Royal Statistical Society: Series B*, 2025.
- [28] Yuqing Liang, Jiancheng Xiao, Wensheng Gan, and Philip S. Yu. Watermarking techniques for large language models: A survey. *Artificial Intelligence Review*, 2026.
- [29] Aaron Lou, Chenlin Meng, and Stefano Ermon. Discrete diffusion modeling by estimating the ratios of the data distribution. In *ICML*, 2024.
- [30] Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. Large language diffusion models. In *NeurIPS*, 2025.
- [31] Jingyang Ou, Shen Nie, Kaiwen Xue, Fengqi Zhu, Jiacheng Sun, Zhenguo Li, and Chongxuan Li. Your absorbing discrete diffusion secretly models the conditional distributions of clean data. In *ICLR*, 2025.
- [32] Fabien A. P. Petitcolas, Ross J. Anderson, and Markus G. Kuhn. Information hiding — a survey. In *Proceedings of the IEEE*, 1999.
- [33] Ohad Raban, Ethan Fetaya, and Gal Chechik. LR-DWM: Efficient watermarking for diffusion language models. *arXiv preprint arXiv:2601.12376*, 2026.
- [34] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. Technical report, OpenAI, 2019.
- [35] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.
- [36] Subham Sekhar Sahoo, Marianne Arriola, Aaron Gokaslan, Edgar Marroquin Marroquin, Alexander M. Rush, Yair Schiff, Justin T. Chiu, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *NeurIPS*, 2024.
- [37] Jiaxin Shi, Kehang Han, Zhe Wang, Arnaud Doucet, and Michalis Titsias. Simplified and generalized masked diffusion for discrete data. In *NeurIPS*, 2024.
- [38] Qingyan Wei, Yaojie Zhang, Zhiyuan Liu, Pengfei Zeng, Yu Wang, Biao Qi, Dong Liu, and Linfeng Zhang. Accelerating diffusion large language models with SlowFast sampling: The three golden principles. In *ICLR*, 2026.

- 484 [39] Chengyue Wu, Hao Zhang, Shuchen Xue, Zhijian Liu, Shizhe Diao, Ligeng Zhu, Ping Luo,
485 Song Han, and Enze Xie. Fast-dLLM: Training-free acceleration of diffusion LLM by enabling
486 KV cache and parallel decoding. In *ICLR*, 2026.
- 487 [40] Linyu Wu, Linhao Zhong, Wenjie Qu, Yifan Li, Yue Liu, Shengfang Zhai, Chunhua Shen, and
488 Jiaheng Zhang. DMark: Order-agnostic watermarking for diffusion large language models.
489 *arXiv preprint arXiv:2510.02902*, 2025.
- 490 [41] Jiacheng Ye, Zhihui Xie, Lin Zheng, Jiahui Gao, Zirui Wu, Xin Jiang, Zhenguo Li, and Lingpeng
491 Kong. Dream 7B: Diffusion large language models. *arXiv preprint arXiv:2508.15487*, 2025.
- 492 [42] Xuandong Zhao, Yu-Xiang Wang, and Lei Li. Protecting language generation models via
493 invisible watermarking. In *ICML*, 2023.
- 494 [43] Xuandong Zhao, Prabhanjan V. Ananth, Lei Li, and Yu-Xiang Wang. Provable robust water-
495 marking for AI-generated text. In *ICLR*, 2024.
- 496 [44] Fengqi Zhu, Rongzhen Wang, Shen Nie, Xiaolu Zhang, Chunwei Wu, Jun Hu, Jun Zhou, Jianfei
497 Chen, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. LLaDA 1.5: Variance-reduced preference
498 optimization for large language diffusion models. *arXiv preprint arXiv:2505.19223*, 2025.